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ORIGINAL ARTICLE

Is the settlement of a winter territory in the European Robin *Erithacus rubecula* triggered by the arrival of conspecific migrants?

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Abstract
European Robins are well known for defending a winter territory. Both resident birds and newly arrived migrants defend a winter territory. However, little is known about the timing and drivers of the settlement of the winter territory in autumn. We hypothesized that settlement is triggered by the arrival of migratory Robins from the North. The hypothesis was tested by relating observational data on first autumn song (as indicator for the settlement of a winter territory) of European Robins in a back yard in the North of the Netherlands to nationwide arrival dates of migratory European Robins in the Netherlands. Date of first autumn song in the studied yard advanced significantly with 14 days from 1993 to 2021, but there was no significant relationship with nationwide autumn arrival dates. We conclude that first autumn song (as indicator of settlement of winter territory) in European Robin does not seem to be triggered by the arrival of (conspecific) migrants. We suggest instead that advanced timing of breeding caused the advancement of winter territory settlement. Climate change may allow resident Robins to moult earlier and settle before conspecific migrants arrive. However, this suggestion needs further testing.

Keywords European Robin · Winter territory · Migration timing · Climate change

Zusammenfassung
Hängt das Besetzen eines Winterreviers beim Rotkehlchen *Erithacus rubecula* von der Ankunft ziehender Artgenossen ab?

Sowohl sesshafte als auch neueintreffende ziehende Rotkehlchen sind für die Verteidigung ihrer Winterquartiere bekannt. Über den Zeitpunkt der und die Ursachen für das Besetzen solcher Winterreviere im Herbst ist jedoch wenig bekannt. Wir stellten die Hypothese auf, dass die Revierbesetzung von der Ankunft der ziehenden Rotkehlchen aus dem Norden abhängt. Die Hypothese wurde getestet, indem Beobachtungsdaten zum ersten Herbstgesang (als Indikator für das Besetzen eines Winterreviers) von Rotkehlchen auf einem Gartengrundstück im Norden der Niederlande in Beziehung zu landesweiten Ankunftsdaten von durchziehenden Rotkehlchen in den Niederlanden gesetzt wurden. Das Datum des ersten Herbstgesangs im untersuchten Garten verfrühte sich von 1993 bis 2021 um 14 Tage. Es gab jedoch keinen signifikanten Zusammenhang mit den landesweiten Herbstankunftsdaten. Wir schließen daraus, dass der erste Herbstgesang (als Indikator für das Besetzen eines Winterreviers) beim Rotkehlchen offenbar nicht durch die Ankunft von ziehenden Artgenossen verursacht wird. Wir gehen stattdessen davon aus, dass ein verfrühter Brutzeitpunkt zu einer früheren Besetzung eines Winterreviers führt. Der Klimawandel könnte es sesshaften Rotkehlchen ermöglichen, früher zu mausern und sich somit in einem Revier niederzulassen, bevor ziehende Artgenossen eintreffen. Diese Vermutung muss jedoch weiter geprüft werden.

Introduction

Many bird species are known to be territorial in winter (cited in Cuadrado 1997), but our understanding of the functions and mechanisms of winter territoriality is limited. One songbird species of the Western Palearctic that is well known for defending a winter territory is the European Robin (Lack 1953; Cramp 1988). However, only little is known about

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the ecology of winter territories in European Robins (e.g. Cuadrado 1997; Lack 2016).

In general, a territory is functional for several factors of the life-history in vertebrate species, like food and non-food resources, population density or mates (Maher and Lott 2000). The actual function of winter territories in European Robin however has been disputed in the past. Despite exhaustive study of Robins in the UK, Lack (1953) could not convincingly proof any function. He suggested that the territorial behaviour in autumn may be caused by a partial revival of the spring breeding behaviour, due to a partial recurrence of the physiological state of early spring, as is shown in some other species (Lack 1953). However, this explanation is not totally convincing because female Robins also have winter territories. Adriaensen and Dhondt (1990) concluded that good access to food is an important function of the winter territory for Robins. On the other hand, Cuadrado (1997) concluded from a study in the Mediterranean that good refuge and shelter, as an anti-predator strategy, may be the reason for winter territories.

Thus, the function of the winter territory in European Robins is still not clear. In addition, other aspects of winter territoriality in Robins are poorly known, e.g., timing and drivers of settlement of the winter territory. The settlement of the winter territory starts in late summer/early autumn, depending on the geographical latitude of the location (Cramp 1988). In the Netherlands, the European Robin is mainly a resident bird, but the resident population is being supplemented every winter with migrants from East and Northeastern-Europe (www.vogeltrekatlas.nl). Both resident birds and newly arriving immigrants settle a winter territory (Lack 1953; Cramp 1988). The migrants arrive from late summer onwards, from August to November, (www.vogeltrekatlas.nl) in the Netherlands. In European Robins, the settlement in a (winter or breeding) territory is strongly associated with singing to defend the territory (Lack 1953). Both males and females defend a winter territory by singing.

In this study, we investigate the relationship between winter territory settlement, indicated by the first autumn song to defend the territory in a studied yard in the North of the Netherlands, and the arrival of migrating Robins across the country for a 29-year period (1993–2021).

Materials and methods

Study area

The studied yard is located in a rural landscape in the province of Groningen, the Netherlands (53° 16', 6° 19'). It has an area of one ha, containing a varied habitat of flower and vegetable garden with lawns, dispersed trees and bushes of different ages (Fig. 1). It is surrounded by woodlots and broad hedges with partly old trees. The Robins mostly sing in top of the trees and bushes and in the surrounding woodlots and hedges. They also use the lawns and gardens for foraging. Such a site is considered as a good winter habitat for Robins in The Netherlands (Bijlsma 2018). Every autumn, 3–5 Robins settled a territory in the yard during the study period.

Fig. 1 Impression of the habitat of 3–5 territories of European Robin in the studied yard during 1993–2021. This image was captured on November 17, 2019 (photo E. B. Oosterveld)

Observation practice

Lack (1953) points out that there is a strong relation between song and occupied territory, both in spring and winter. Following Lack (1953), we assume that a singing bird is defending a territory. Dutch European Robins are mainly residents and during winter, these residents are being supplemented by arriving migrants from Northeastern and Eastern Europe. Both residents and migrants sing to defend their winter territory. Whether it is a resident bird that sings, or a newly arrived migrant is irrelevant for testing our hypothesis as we relate the first song as indication for territory settlement to the national arrival dates of the migratory birds.

The territory settlement starts with one bird on day x starting to sing, with more birds following the next days. Once settled, the birds frequently sing throughout the day. Therefore, we usually need no long observation time to recognize territory settlement. The birds often sing against each other, which makes quickly clear where exactly the territories are in the yard. With this information, it is easy to identify the number of individual territory holders in the studied yard. Being present at the yard nearly every day, the first author determined and registered the exact day of the first song every year from 1993 to 2021.

Arrival of migrants

To study the relationship of the first song with the arrival of migrants we compared the yearly date of first song with the yearly data on autumn migration of the Robin across the Netherlands extracted from www.trektellen.nl. Since we lacked information of local arrival of migrants at the studied yard, we used the migration data on a national scale as a proxy of migrant arrival at the studied yard. The migration data on a national level are representative for the arrival of migratory birds at the studied yard as is shown in Fig. S1. The figure presents distribution maps of migrating Robins across the country during September, the month of first song at the studied yard, for the studied period. The distribution maps show that during the first half as well as the second half of September migrating Robins are observed all across the country, including south of the studied location.

Furthermore, in Figure S2 we show an increase of captured Robins during the autumns of 2009–2021 at a mist netting site in another backyard at 9.5 km from the studied backyard. Birds are captured there throughout the year for ringing purpose. The increase in captured Robins from August onwards is a result of the arrival of conspecific migrants in the area.

The trektellen.nl website reports migration data of many bird species at thousands of locations all over the United States and Europe. In the Netherlands, counts on hundreds of locations all across the country are reported every year. The average number of birds per hour per species over the counting sites are reported every week. In our study, we only used data with a minimum of ten locations across the country per year. The migration data reflect the arrival and passage of the birds in and over the Netherlands. We presume that the migration data indicate both passage and arrival in the winter quarters at the same time.

To test whether the start of singing is related to the arrival of migrants, we compared the yearly dates of first song in the studied yard with (1) the first week of migrants arriving, (2) the arrival of the first 10% of the migrants (10th percentile) and (3) the median week of the yearly autumn migration (Fig. 2).

Analysis

We used a general linear model with the Julian date of the first song (FS) in the yard as dependent variable and year (Y) as well as the median (M), 10th percentile (P) and first week (F) of arriving Robins in the Netherlands as fixed factors (see Table 1 for the model equation). We run the general linear model with the R-package lme4 (Bates et al. 2015). Before running the model, we checked for potential correlation between the independent variables, which was absent (Fig. S3), and visualized it with the R-package corrplot (Wei & Simko 2021). If a correlation would have been larger than 0.7, one of the two variables would have been needed to be excluded from analysis (Zuur et al. 2009). Visualization of the figures was done with the R-packages ggplot2 (Wickham 2016), ggeffects (Lüdecke 2018) and patchwork (Pedersen 2022). The model was validated by checking for normality, homogeneity and independence (Zuur et al. 2009). Model validation results show homogeneity (Fig. S4a), normality (Fig. S4b) and independence (Fig. S4c–f) of the data. All analysis were conducted in R (version 4.2.2) (R Development Core Team 2023).

Results

According to a significant linear relationship between year and date of first song, the first song of the Robins at the studied yard advanced from September 21 to September 7, i.e. 14 days during the study period (Fig. 3a & Table 1; t-value = -2.45, p = 0.03, df = 11).

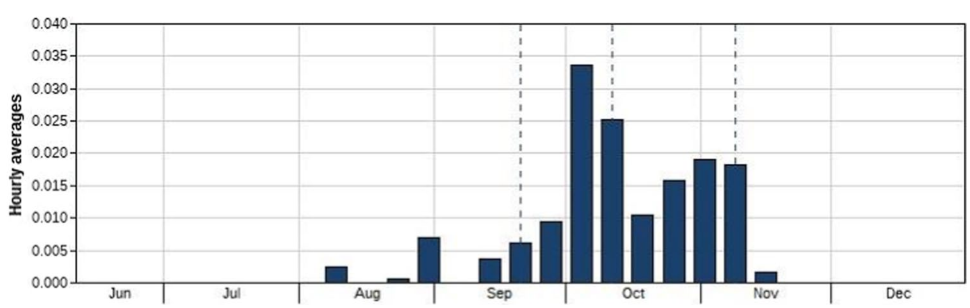

Fig. 2 Average number of weekly reported robins across the Netherlands (shown for 2021 as an example). Data were extracted from www.trektellen.nl. The total number of observed robins is 264, the total number of observation hours equals 25.616. First arrival of robins is early August, thereafter increasing recordings are made, the left dashed line is the week of the 10-percentile of the arriving birds, the middle dashed line is the week of the median of the arriving birds. September 1 is day 244 in week 34

Table 1 Model results showing the regression coefficients, standard error, t-value and p- value per variable ($n = 16$)

Variable	Coefficients	Standard error	*t*-value	*p*-value
Intercept	**1421.59**	**48,510**	**2.93**	**0.01**
Year	**– 0.58**	**0.24**	**– 2.45**	**0.03**
First week arriving	– 1.67	1.09	– 1.53	0.16
10th percentile arriving	0.50	1.04	0.48	0.64
Median arriving	0.78	1.44	0.55	0.60

Bold rows indicate significant ($p < 0.05$) variables. The model equation is $FS_i = \beta_0 + \beta_1 Y_i + \beta_2 M_i + \beta_3 P_i + \beta_4 F_i + \varepsilon_i$ FS first song, *Y* year, M median week, *P* week of 10 percentile, *F* first week

β_0 represents the intercept and $\beta_{1\text{-}4}$ the regression coefficients of the model. i represents each observation. The unexplained information is captured by the residuals ε, which are assumed to be normally distributed with expectation 0 and variance σ^2. Due to missing data among different variables, the total sample size of the model resulted in 16 data points (i)

We found no relation between the date of first song and the first week at which autumn migration of the Robins in the Netherlands started (Fig. 3b & Table 1; t-value $= -1.53$, $p = 0.16$, $df = 11$). In addition, we did not find a relationship between the first song at the studied yard and the first 10th percentile of arriving migratory birds (Fig. 3c & Table 1; t-value $= 0.48$, $p = 0.64$, $df = 11$), or the median week of arriving migratory birds (Fig. 3d & Table 1; t-value $= 0.55$, $p = 0.60$, $df = 11$). The model explained 35% of the variation in the timing of Robin first autumn song ($R_{adj} = 0.3511$).

Discussion

Our study shows two main outcomes: (1) over the last 29 years, the date of first autumn song at the studied yard advanced by two weeks and (2) there is no relationship between the date of first autumn song and the arrival of conspecific migrants from Eastern and Northeastern Europe to the Netherlands.

In fact, the analysis shows a nearly significant relationship ($p = 0.16$) between the first song at the studied yard and the first week of arrival of migratory Robins in the Netherlands. This suggests that more data points in a longer time series could result in a significant relationship. On the other hand, a time series of 20 data points over 29 years is a large data set, which is quite exceptional in ecological research and that should already show robustness of the presented relationship. Nevertheless, we recommend longer time series and more study sites to draw more large-scale and robust conclusions in the future.

Thus, the advancement of the date of first autumn song at the yard does not seem to be caused by a potential advanced arrival of migrants from their breeding grounds in Eastern and Northeastern Europe. In fact, due to climate change we would rather expect that migrants stay nowadays longer on more northern breeding grounds than in the past. One possible explanation of a lack of relation would be that first arrival of migrants on a national scale is not a good indicator for the settlement (followed by first song) of the migrants in the studied location. That would be for instance the case if the first migrants would all fly by the studied yard to their wintering grounds further away. To overcome this potential missing effect, we included not only the very first week but also the week of the 10th percentile of the migrating birds and the median week of the yearly autumn migration. It seems likely that the latter two variables would include the first Robins that settle their winter territories in the country and that this timing would result in a significant relationship between arrival of migratory birds and date of the first song, if there would be a relationship.

However, what may explain the advancement of the settlement of the winter territories in the studied location? We suggest that the advancement of the settlement of the species is caused by climate change. In the period from 1986 to 2015, songbirds in the Netherlands advanced their egg laying date with eight days, likely because of advanced insect availability for their chicks due to warmer springs (CBS et al. 2017, compare Dunn and Winkler 2010). The CBS et al. analysis is based on data of a total of 44 breeding songbirds in the Netherlands, including the European Robin. Unfortunately, there are no separate data of the European Robin presented in this study. Directly after fledging, Robins moult their feathers over 50–60 days (Ginn and Melville 1983), during which they do not sing (Cramp 1988). The earlier they breed, the earlier they can finish moulting, being ready to resume singing and settle in a winter territory. Earlier breeding as a cause of earlier settlement of a winter territory is also in line with the observation that young birds start singing in autumn before adults do (Lack 1953).

A relationship of first autumn song with advanced timing of breeding would imply that resident breeding birds are the first to settle, or perhaps to continue their spring territories. This is emphasized by the lack of a relationship between autumn first song and the timing of arriving migratory birds in this present study. Migrants may additionally settle over autumn, also defending their territory by singing (Lack 1953). Being first with territory settlement may be beneficial for the resident birds as they can choose the high-quality territories.

A relationship between first autumn song with advanced timing of breeding would also imply that Robins do not react on climate change by extending the breeding period. European Robins usually breed twice at temperate climate zones (Lack 1953; Cramp 1988). And it has been shown that as a reaction on climate change, especially multi-brooded species extend the breeding period with advancing breeding

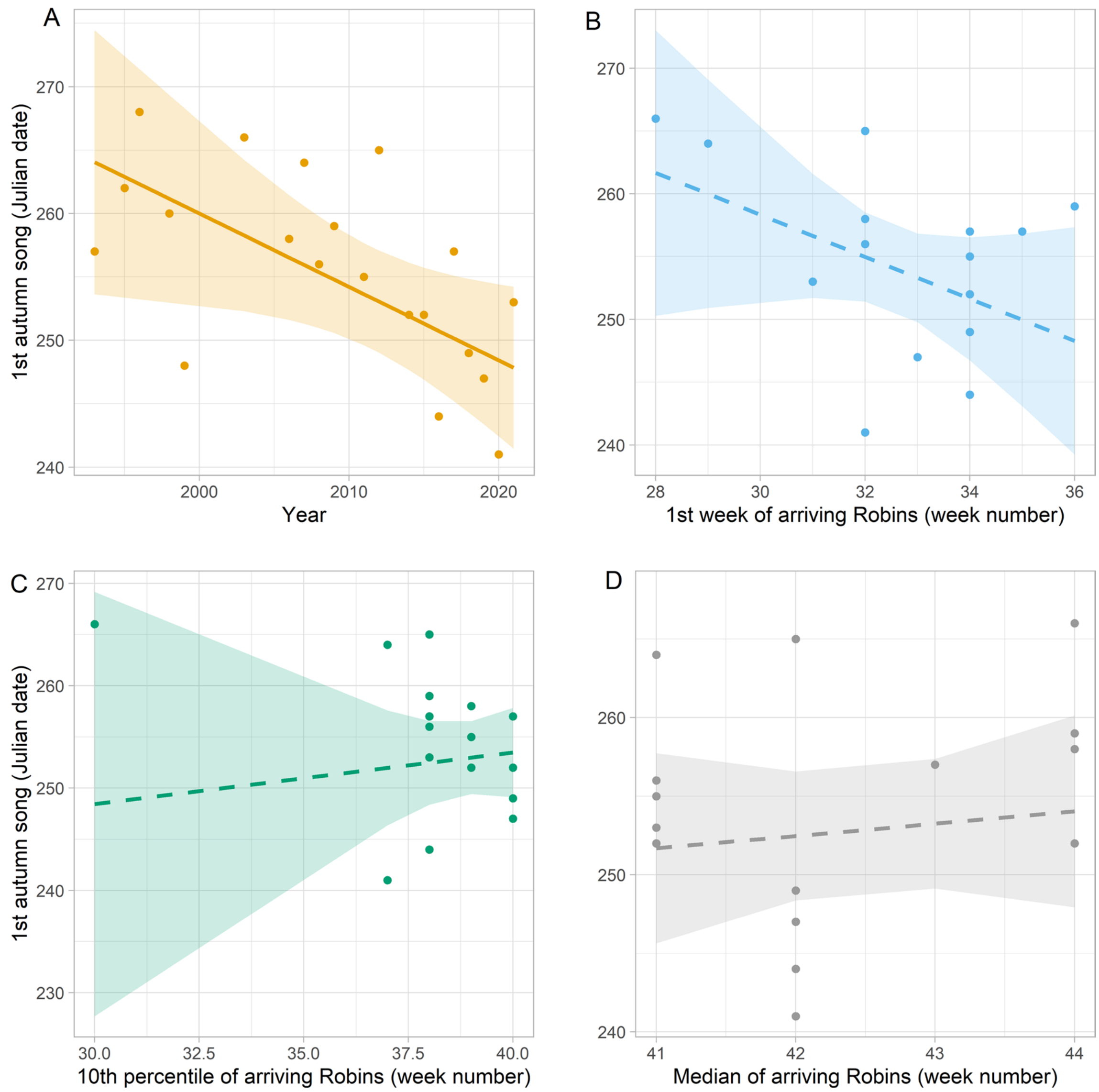

Fig. 3 Relationship between the Julian date of first autumn song of European Robins at the studied yard and **a** year, **b** the first week, **c** the first 10th percentile and **d** the median week of Robins arriving in the Netherlands at autumn migration. Solid and dashed lines indicate significant ($p<0.05$) and non-significant ($p>0.05$) relationship, respectively. Shaded ribbons represent the 95% confidence interval and dots represent the measured data points. 240 Julian date being 28th August in regular years

dates of first clutches and delaying breeding dates of second clutches (Dunn and Møller 2014). For example, Barn Swallows *Hirundo rustica* have advanced the laying date of first clutches and delayed (or advanced less) the laying date of second clutches, by extending the interval between first and second clutches (Møller 2007; Altenburg et al. 2022). Consequently, parents with longer intervals between clutches were able to rear more offspring, and females with two clutches per year survived better when the interval between clutches was long (Møller 2007). To our knowledge, there is no information on extended breeding period in European Robins.

Conclusions

We found no relationship between the first date of autumn song of European Robin in the studied location, as an indication of the start of settlement of the winter territory, and the first arrival of eastern and northeastern migratory Robins in the Netherlands, nor with the date of the 10th percentile and the median date of the arrival of migratory Robins. Therefore, we conclude that first autumn song in the studied yard is not related to the arrival of migrating Robins. The striking advancement of the first autumn song by 14 days during the study period of 29 years is likely better explained by an advancement of the breeding period due to climate change.

This advancement of breeding may enable the resident Robins to moult earlier after the breeding season and start settling earlier their winter territory. Thus, earlier breeding enables the resident Robins to occupy a winter territory of high quality before the migratory conspecifics arrive. However, this new developed hypothesis needs further testing.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10336-024-02150-7.

Data availability Data files and R code to reproduce the results will be made openly available on Github upon acceptance of the manuscript.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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